NLP1 Tutorial 1: Quick Guide

Nov 9, 2026 · @Isha

The tutorial builds a text classifier in five steps: Unicode text → BPE tokens → a vector → softmax regression or an MLP → a Bayesian comparison of systems. Below: every concept in one line, then every tutorial question with a one-line answer. The full version is NLP1 Tutorial 1: Concepts and Questions.

Concepts in one line

Concept	In one line
Unicode / code point	one big table numbering every character (é = U+00E9)
Encoding (ASCII, Latin-1, UTF-8)	how code points become bytes; UTF-8 uses 1–4 bytes and covers everything
Wrong encoding	an error, or silent garbage ("CafÃ©", called mojibake)
Orthographic word	a string between spaces; Chinese has none
Morpheme / listeme	smallest meaningful part (cat + -s) / anything memorised whole ("kick the bucket")
Heaps' law	$V(n) = K \cdot n^{\beta}$: the vocabulary keeps growing with more text ($\beta \approx 0.4$–0.6 for English)
BPE (fitting)	start from characters, repeatedly merge the most frequent neighbouring pair, record the order
BPE (tokenizing)	pre-tokenize, then replay the merges in the learned order; unseen characters → unknown
BPE settings	vocab size (more merges = longer units), min frequency, pre-tokenizer (what can never merge)
Classifier	probabilistic model (text → pmf over K classes) + decision rule (pmf → one class)
Most probable class	the usual decision rule: argmax of the pmf; a convention, not a law
Tabular CPD	one row of probabilities per possible text: infinite, sparse, can't handle new text

Concept	In one line
Encoder / named axes	turns text into a fixed-length vector; each dimension means "token v occurs" or "how often"
Bag-of-words / bigrams	counts per token (order lost) / counts per neighbouring pair (local order kept)
Softmax regression	score_k = w_k·h(x) + b_k, softmax turns scores into probabilities; convex
MLP	linear → ReLU → linear → softmax; can model combinations of words
LogSoftmax + NLLLoss	= cross-entropy, computed stably
Cross-validation	k folds, train on k−1, test on 1, repeat; seeds = repeated runs
NHST vs estimation	a p-value tests "exactly zero"; estimation shows how big a difference plausibly is
Correlated t-test	Bayesian posterior over the mean difference of fold scores, variance inflated because folds share data (ρ = 1/k)
ROPE	differences too small to matter (tutorial: 0.005 accuracy)
P(A) / P(tie) / P(B)	posterior area left of / inside / right of the ROPE
HDI	the narrowest range holding 95% of the posterior
Effect size	posterior mean difference / posterior standard deviation
Ensembles	majority vote; mixture of experts (average pmfs); product of experts (average logits)
devtest	an extra split for choices like ensembling, so the test set stays untouched

Every tutorial question, one line

Question	Answer
Segmentation: characters as units for European languages?	Each character carries little meaning, sequences get long, models must build meaning from scratch.
English word rules for other languages?	Only partly: German compounds, French clitics (l'homme), Finnish suffixes need their own rules.
Similar vocabulary size across languages?	No: morphologically rich languages produce many more word forms.
BPE: effect of vocab size / merges / min frequency?	More merges = longer, more word-like units; higher min frequency = fewer merges of rare pairs.
Effect on informativeness?	Whole words carry more meaning, but very rare units get poorly learned weights.
Lowercasing and pre-tokenization?	Lowercasing frees slots but loses case cues; pre-tokenization decides what can never merge.
Classification: parameters of a literal table?	One row per possible text × (K−1): infinitely many.
Sparse or dense table?	Very sparse: nearly every text occurs once or never.
Classify a new text with the table?	No: there is no row for it.
Why tables don't scale for language?	Language is productive; tables can't share what they learned between similar texts.
When does argmax go against intuition?	Near-flat pmfs (0.34/0.33/0.33), or unequal costs (missing a tumour).
Representation: dimension of a bag-of-words vector?	V, the vocabulary size.
Does it depend on the text's length?	No; length changes the values, not the dimension.
Man bites dog vs dog bites man?	Same vector: order is lost.
With a BPE tokenizer?	Still the same: it is still a bag.

Question	Answer
With bigrams?	Up to V^2 dimensions, still fixed; now the two sentences differ.
Log-linear: why can it score unseen text?	Parameters belong to shared features, not to whole texts.
Why can the best parameters still give bad pmfs?	The representation may not separate the texts ("not good" vs "good"), and the model is linear.
Experiment: is accuracy good for AG News?	Yes: its 4 classes are balanced.
An alternative?	Macro-F1, balanced accuracy, per-class precision/recall.
Why might counts beat indicators?	A repeated topic word signals the topic more strongly.
Final 1: Unicode characters as units?	Tiny vocabulary, no unknowns, but almost no meaning per unit.
Final 2: is a larger vocabulary always better?	No: rarer features, more weights, diminishing returns.
Final 3: linguistic tokenizer (NLTK) better?	Whole words, but an open, ever-growing vocabulary with unknown words; empirical question.
Final 4: tokenizer trained on Europarl, classifier on news?	Domain mismatch: news words get split into small, less informative pieces.
Final 5: two representation ideas?	TF-IDF (we did: +0.02–0.03 for log-linear) and bigrams.
Final 6: one architecture idea?	A second hidden layer (we did: matches the MLP after a redesign), or a class-weighted loss.
Final 7: defend or challenge accuracy with your data?	Look at the label distribution: our professor share is 30%, so use macro-F1.